Results

ReactionTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	7294	2.000	3647	13.09	< .001	0.5925
	Greenhouse-Geisser	7294	1.400	5209	13.09	.002	0.5925
Residuals	None	5016	18.00	278.7			
	Greenhouse-Geisser	5016	12.60	398.0			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$1.152\times10^{+4}$	9	1280		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	201.6	21.15	6.688	0.1049
go/no-go	10	216.6	29.47	9.318	0.1361
go/stop	10	239.5	22.84	7.223	0.09537

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.5718	4.472	2	.107	0.7002	0.7898	0.5000

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−15.00	6.036	9	−2.485	−0.6060	.069
	go/stop	−37.92	6.212	9	−6.105	−1.532	< .001
go/no-go	go/stop	−22.92	9.601	9	−2.387	−0.9262	.069

Note. P-value adjusted for comparing a family of 3 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	16.20	2	< .001	0.8100

MovementTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	$1.529\times10^{+5}$	3.000	$5.095\times10^{+4}$	38.32	< .001	0.8098
	Greenhouse-Geisser	$1.529\times10^{+5}$	1.895	$8.064\times10^{+4}$	38.32	< .001	0.8098
Residuals	None	$3.590\times10^{+4}$	27.00	1329			
	Greenhouse-Geisser	$3.590\times10^{+4}$	17.06	2104			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$9.566\times10^{+4}$	9	$1.063\times10^{+4}$		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	556.2	30.37	9.605	0.05461
go/no-go	10	589.3	66.04	20.88	0.1121
go/stop	10	612.8	50.66	16.02	0.08267
free	10	721.1	82.26	26.01	0.1141

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.2897	9.568	5	.091	0.6318	0.7955	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−33.10	14.02	9	−2.360	−0.5476	.085
	go/stop	−56.53	9.569	9	−5.908	−0.9352	< .001
	free	−164.9	20.60	9	−8.005	−2.728	< .001
go/no-go	go/stop	−23.43	14.06	9	−1.667	−0.3876	.130
	free	−131.8	21.77	9	−6.053	−2.180	< .001
go/stop	free	−108.4	14.53	9	−7.458	−1.793	< .001

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	25.68	3	< .001	0.8560

PeakVelTop_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	0.7174	3.000	0.2391	0.9784	.417	0.09805
	Greenhouse-Geisser	0.7174	2.265	0.3167	0.9784	.402	0.09805
Residuals	None	6.600	27.00	0.2444			
	Greenhouse-Geisser	6.600	20.39	0.3237			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	178.5	9	19.83		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	31.33	2.033	0.6430	0.06490
go/no-go	10	31.45	2.378	0.7519	0.07559
go/stop	10	31.70	2.371	0.7497	0.07479
free	10	31.47	2.271	0.7182	0.07217

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.5577	4.510	5	.482	0.7551	1.000	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−0.1255	0.2784	9	−0.4507	−0.05534	1.000
	go/stop	−0.3711	0.2123	9	−1.748	−0.1637	.687
	free	−0.1400	0.2145	9	−0.6526	−0.06173	1.000
go/no-go	go/stop	−0.2456	0.1535	9	−1.600	−0.1083	.721
	free	−0.01448	0.2462	9	−0.05882	−0.006387	1.000
go/stop	free	0.2311	0.2014	9	1.148	0.1019	1.000

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	1.560	3	.668	0.05200

PeakFz_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	0.04809	3.000	0.01603	4.339	.013	0.3253
	Greenhouse-Geisser	0.04809	2.279	0.02110	4.339	.023	0.3253
Residuals	None	0.09975	27.00	0.003694			
	Greenhouse-Geisser	0.09975	20.51	0.004864			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	0.5840	9	0.06489		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	1.234	0.1294	0.04091	0.1048
go/no-go	10	1.242	0.1236	0.03908	0.09948
go/stop	10	1.272	0.1461	0.04620	0.1149
free	10	1.323	0.1504	0.04755	0.1137

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.5532	4.571	5	.474	0.7596	1.000	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−0.008350	0.01957	9	−0.4267	−0.06059	.778
	go/stop	−0.03782	0.03142	9	−1.204	−0.2745	.778
	free	−0.08860	0.03376	9	−2.624	−0.6429	.138
go/no-go	go/stop	−0.02947	0.02542	9	−1.160	−0.2139	.778
	free	−0.08025	0.02432	9	−3.299	−0.5823	.055
go/stop	free	−0.05077	0.02619	9	−1.939	−0.3684	.338

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	8.400	3	.038	0.2800

HookTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	$1.320\times10^{+5}$	3.000	$4.402\times10^{+4}$	28.06	< .001	0.7571
	Greenhouse-Geisser	$1.320\times10^{+5}$	1.797	$7.348\times10^{+4}$	28.06	< .001	0.7571
Residuals	None	$4.236\times10^{+4}$	27.00	1569			
	Greenhouse-Geisser	$4.236\times10^{+4}$	16.17	2619			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$2.389\times10^{+5}$	9	$2.655\times10^{+4}$		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	272.7	79.96	25.29	0.2933
go/no-go	10	307.6	88.72	28.05	0.2884
go/stop	10	352.6	80.16	25.35	0.2273
free	10	426.4	102.8	32.50	0.2411

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.3432	8.258	5	.145	0.5990	0.7391	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−34.95	12.78	9	−2.734	−0.3954	.031
	go/stop	−79.95	17.46	9	−4.579	−0.9044	.004
	free	−153.7	22.77	9	−6.751	−1.739	< .001
go/no-go	go/stop	−45.00	15.13	9	−2.974	−0.5090	.031
	free	−118.8	22.41	9	−5.300	−1.344	.002
go/stop	free	−73.77	12.83	9	−5.749	−0.8346	.001

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	25.44	3	< .001	0.8480

LoopTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	4688	3.000	1563	4.711	.009	0.3436
	Greenhouse-Geisser	4688	1.715	2734	4.711	.030	0.3436
Residuals	None	8956	27.00	331.7			
	Greenhouse-Geisser	8956	15.43	580.4			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$7.012\times10^{+4}$	9	7791		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	216.9	58.11	18.38	0.2680
go/no-go	10	212.9	43.49	13.75	0.2043
go/stop	10	195.1	39.62	12.53	0.2031
free	10	224.6	44.14	13.96	0.1965

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.3212	8.771	5	.121	0.5715	0.6930	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	3.993	8.283	9	0.4821	0.08521	.944
	go/stop	21.76	11.28	9	1.929	0.4643	.257
	free	−7.763	10.34	9	−0.7511	−0.1656	.944
go/no-go	go/stop	17.77	6.266	9	2.836	0.3791	.098
	free	−11.76	5.074	9	−2.317	−0.2508	.183
go/stop	free	−29.52	5.508	9	−5.360	−0.6299	.003

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	10.92	3	.012	0.3640

AccelTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	145.1[a]	3.000[a]	48.36[a]	2.676[a]	.067[a]	0.2292
	Greenhouse-Geisser	145.1	1.773	81.84	2.676	.104	0.2292
Residuals	None	487.9	27.00	18.07			
	Greenhouse-Geisser	487.9	15.95	30.58			

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated ($p < .05$).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	3381	9	375.7		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
simple	10	66.11	11.07	3.499	0.1674
go/no-go	10	68.30	9.927	3.139	0.1453
go/stop	10	64.77	9.027	2.855	0.1394
free	10	69.69	11.29	3.569	0.1620

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.1615	14.08	5	.016	0.5908	0.7253	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
simple	go/no-go	−2.188	2.304	9	−0.9496	−0.2110	.734
	go/stop	1.348	2.330	9	0.5784	0.1300	.734
	free	−3.574	2.449	9	−1.460	−0.3448	.535
go/no-go	go/stop	3.536	1.220	9	2.898	0.3411	.103
	free	−1.386	0.7817	9	−1.774	−0.1337	.440
go/stop	free	−4.922	1.688	9	−2.916	−0.4748	.103

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	11.30	3	.010	0.3768